

Problem 2 (25 points). Semitic semi-Seminar on Cognates

Arabic and Hebrew² today are two different, mutually unintelligible languages, but apart from grammatical similarities they do share a number of lexical correspondences as well. Besides loanwords (mostly from Arabic to Hebrew) from different historical periods, scholars identified over a thousand *cognates*: words with a common etymological origin from Proto-Semitic, the partially reconstructed common ancestor language spoken around 6000 years ago.

The two lists below show pairs of cognates, but the pairs are mixed: Arabic on the left, Hebrew on the right. The first match (1- a) is given for you.

شمس	1.	←————→	a.	שמש
غضب	2.		b.	כלב
ولد	3.		c.	מלך
أرض	4.		d.	עבד
بطن	5.		e.	ארץ
كلب	6.		f.	עצב
حبل	7.		g.	קרן
ملك	8.		h.	ילד
قرن	9.		i.	חבל
عبد	10.		j.	בטן

Thanks to regular and consistent sound changes, we have some easily identifiable patterns. Take for example the following words from the list above (transcribed into the Latin script):

Arabic	Hebrew	Translation	Arabic	Hebrew	Translation
kalb	kelev	dog	shams	shemesh	sun
malik	melekh	king	qarn	qeren	horn
'arḍ	'erets	land, earth	ghaḍab	'etsev	anger, sadness
'abd	'eved	slave	walad	yeled	child

² Arabic and Hebrew are two languages that are spoken around the Middle East and North Africa. Both languages have long histories, going back to ancient times. They belong to the Semitic branch of the Semito-Hamitic language family, which is the old name for what we nowadays call the Afroasiatic language family. Some of the oldest recorded and attested languages belong to this family, for example Egyptian and Sumerian – now dead languages. In this puzzle we are going to look at Modern Standard Arabic and Modern Hebrew, and some parallels between them.

(a) What is the transliteration of the two word-pairs in the lists that we did not give in the examples above?

- a. _____ → _____
b. _____ → _____

(b) The chart below shows a few letters in both scripts, with their transliteration. Note that some letters may have different forms depending on the context they appear in. Fill in the gaps (1) - (15):

Hebrew:

א	ב / בּ	ג	ד / דּ	ה	ו	ז	ח	ט	י	כ	ל
(1)	(2)	(3)	(4)	k	(5)	t	h	(6)	(7)	b	' (alef)

Arabic:

ط / ظ / ط	ر / ر	د / د	ح / ح / ح	ب / ب / ب	أ
ṭ	(8)	(9)	ḥ	(10)	' (alif)

و / و	ن / ن / ن	ل / ل / ل	ك / ك / ك	ض
(11)	(12)	(13)	(14)	(15)

(c) Pair the matching cognates on the first page together.

(d) If *thousand* in Arabic is ألف and the final letter 'f' in Hebrew is פ, what is the transliteration of אלה – also meaning *thousand* in Hebrew?

(!) No additional explanation besides the answers is required, nor will be marked.

 The apostrophe (') in both languages represents a glottal stop /ʔ/. In Arabic it is written by a *hamza*, that often “sits” on top of an *alif*, in Hebrew it is represented by an *alef*, and often omitted in pronunciation. Here, you should just consider it a consonant, and treat it like you would treat any other consonant!

The semi-circle in the top index (ֿ) represents a voiced pharyngeal fricative /ʕ/, which is an odd sound made by contracting the muscles in the throat. It gives Arabic its unique flavor we can easily hear. In Modern Hebrew it is silent and almost only appears in writing. Just think of it as an ordinary consonant!